

OpenStack offers several methods for storing instance data, each presenting different trade-offs between **capacity, performance, and reliability** 1, 2.

Local Storage

While most production environments move toward external solutions, instances can be stored on the **local disk space** of the compute node 1.

- **Characteristics:** The disk space capacity is factored into the sizing of the compute node alongside CPU and RAM 1.
- **Disadvantage:** If the physical compute node fails, recovering the instance data can be difficult because the data resides on that specific piece of hardware 2.

External Shared Storage

In this configuration, instance disks are hosted **externally** and do not reside on the compute nodes themselves 2.

- **Advantages:**
- **Ease of Recovery:** If a compute node fails, the instance can be recovered more easily because its data is safe on the external storage pool 2.
- **Versatility:** The shared pool can be used for other installation purposes beyond just hosting active VM disks 2.
- **Disadvantages/Trade-offs:**
- **Neighbor Noise:** Heavy I/O disk usage by one VM can negatively affect the performance of neighboring VMs sharing the same storage 2.
- **Network Latency:** Because the data must travel over a network, there is a risk of performance degradation compared to local disk speeds 2.

Block Storage (Cinder)

Cinder is the dedicated OpenStack service that provides **persistent block storage volumes** 3, 4.

- **Functionality:** These volumes function like **virtual hard disks** that can be attached to and detached from VMs as needed 3, 5.
- **Access:** Users can manage these volumes through the **Cinder API** or the **Horizon dashboard** 6, 7.

Object Storage (Swift)

While not typically used for the active "boot disk" of a running instance, **Swift** provides an alternative for unstructured data 4, 8.

- **Use Cases:** It is best suited for storing large amounts of **backups, archives, and logs** 8.
- **Scalability:** It offers a highly scalable solution for data that does not require the low-latency performance of block storage 4.

Selection Considerations

When choosing a storage alternative, administrators must evaluate:

- **Performance vs. Capacity:** Local storage may offer lower latency, but external shared storage provides better recovery options and higher total capacity 2.
- **Workload Patterns:** Heavy I/O workloads might require dedicated block storage to avoid the "noisy neighbor" effect seen in shared pools 2.
- **Service Level Agreements (SLAs):** High-availability requirements usually necessitate **shared or block storage** to ensure instances can be rebooted on different nodes if hardware fails 2, 9.

Exam-Ready Notes

- **Cinder (Block Storage):** The standard for **persistent VM disks**; manages volumes like virtual hardware 3, 8.
- **Glance (Image Storage):** Though used for images (templates), it is a central repository that works with Cinder and Nova to deploy instances 5, 10.
- **Shared Storage Benefit:** Primarily **Disaster Recovery**; allows instances to be moved or recovered quickly after a host crash 2.
- **Shared Storage Risk: Network Latency and I/O Bottlenecks** caused by other VMs 2.
- **Swift (Object Storage):** Use for **long-term data**, backups, and unstructured content rather than active OS disks 8.
- **Resource Sizing:** When planning compute nodes, disk capacity must be considered along with CPU/RAM over-commitment ratios 1, 11.